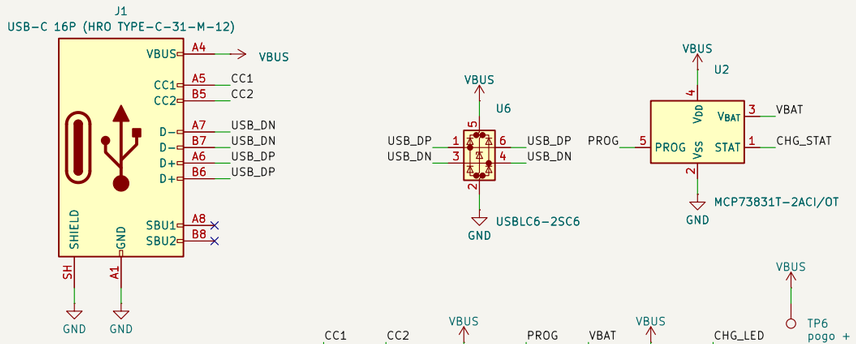
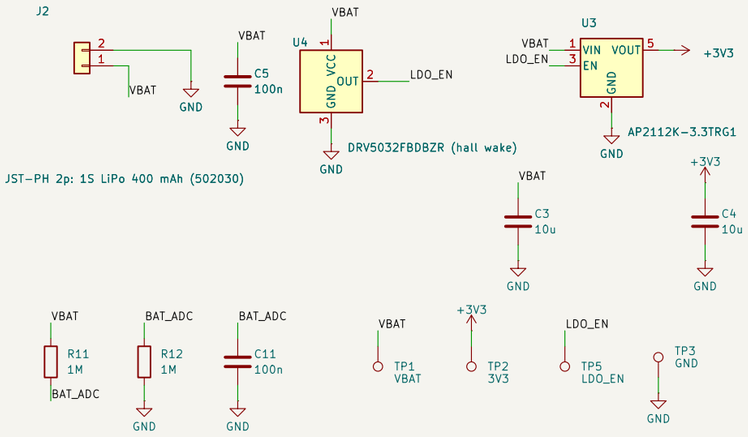
SUTRA relay puck — carrier PCB | ESP32-C3-MINI-1 + RFM95W-868 (SX1276) | 1S 400 mAh, hall-gated power

A. USB-C (charge + native USB flashing), ESD, Li-ion charger



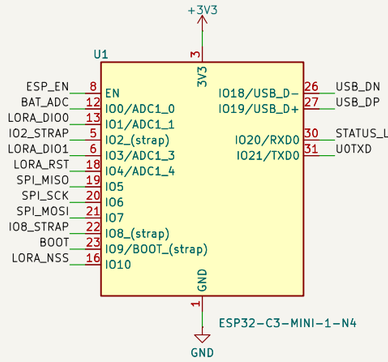
CC1/CC2 5.1k pull-downs = USB-C sink (5 V default current). RPROG 5.1k -> I_REG = 1000 V / 5.1k = 196 mA (0.5 C). LED1 lights while charging (STAT low). TP6/TP7 = 2 mm pogo targets: the cassette can charge pucks without USB.

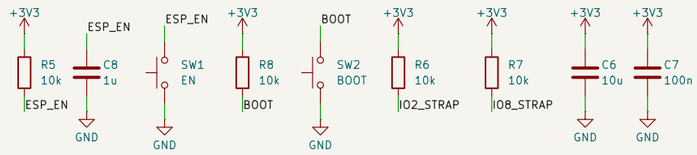
B. 1S LiPo, hall-gated 3.3 V LDO, battery sense

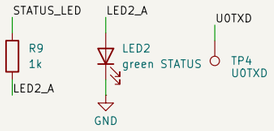


Wake logic: DRV5032FB (omnipolar, push-pull, 5 Hz) runs from VBAT and drives the LDO EN.
Puck in cassette (magnet) -> OUT low -> LDO off, ESP32 unpowered (storage drain ~3 uA total).
Puck leaves the tube -> OUT high -> 3V3 up -> ESP32-C3 boots and joins the relay chain.
VBAT/2 divider -> IO0 (ADC1_CH0): 4.2 V -> 2.1 V, constant 2.1 uA drain.

C. ESP32-C3-MINI-1 (Wi-Fi / ESP-NOW-LR relay + FTM)

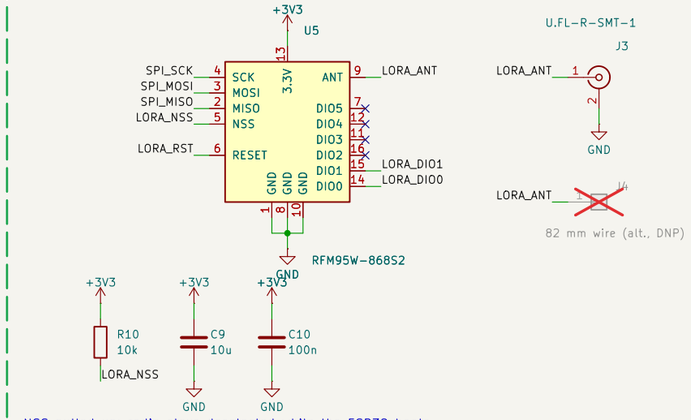






Strapping: IO2 and IO8 held high (10k), IO9 = BOOT (10k up, SW2 to GND) -> SPI boot normally, download mode with SW2. No other load on IO2/8/9.
EN: 10k + 1 uF RC delay after 3V3 rises; SW1 = reset.
IO18/IO19 = native USB-Serial/JTAG: flash + console over the same USB-C.
LoRa SPI through the GPIO matrix (SX1276 max 10 MHz SCK).
DIO0/DIO1 on IO1/IO3 (deep-sleep wake capable, GPIO0-5).
Battery ADC on IO0 = ADC1_CH0 (C3 ADC2 is uncalibrated, erratum: avoid).
IO20 drives the status LED; IO21 (U0TXD) = boot-log test pad.

D. LoRa lifeline — RFM95W-868 (SX1276), 865-867 MHz



NSS pulled up: radio stays deselected while the ESP32 boots.
Antenna: U.FL + 868 MHz pigtail (default) or 82 mm quarter-wave wire on J4 (DNP).
Firmware: 865-867 MHz, PA_BOOST +17 dBm (RFO unused), SF7-SF9/125 kHz, duty <= 2.5 %.

E. Power flags, GPIO map, notes



GPIO map (ESP32-C3-MINI-1)

IO0	BAT_ADC (ADC1_CH0)	IO6	SPI_SCK
IO1	LORA_DIO0 (wake)	IO7	SPI_MOSI
IO2	strap, 10k pull-up	IO8	strap, 10k pull-up
IO3	LORA_DIO1 (wake)	IO9	BOOT button
IO4	LORA_RST	IO10	LORA_NSS
IO5	SPI_MISO	IO18/19	USB D-/D+
IO20	STATUS_LED	IO21	U0TXD test pad

Power: USB-C/pogo 5 V -> MCP73831 -> VBAT (3.0-4.2 V) -> AP2112K-3.3 (600 mA, gated by hall) -> 3V3
Peak 3V3 load ~0.45 A (Wi-Fi TX ~0.35 A + SX1276 TX)
JST-PH polarity is NOT standard across LiPo vendors: check the lead before plugging (no reverse protection).
U1 footprint = ESP32-C6-MINI-1 land pattern (same 13.2 x 16.6 mm LGA-53); verify before layout.

Generated by electronics/scripts/build_puck.py
PCB layout not started (roadmap Month 1)
Gas-free mock-gallery prototype; not intrinsically safe
Vishwakarma Awards 2026-27 Stage 2 design baseline
Team Palanteen, TSEC Mumbai — VK26-AU-PAI-0065

Sheet: /
File: sutra-puck.kicad_sch

Title: SUTRA relay puck — carrier board (ESP32-C3 + SX1276)

Size: A3	Date: 2026-10-22	**Rev: S2-A**
KiCad E.D.A. 10.0.7		Id: 1/1